

Matthew Regehr

DC 2569-200 University Ave W
Waterloo, ON N2L 3G1

matt19234@gmail.com
(825) 461-1185

Education

University of Waterloo <i>PhD in Computer Science</i>	September 2023–Present <i>Waterloo, ON</i>
University of Waterloo <i>Master of Mathematics in Computer Science</i>	September 2021–August 2023 <i>Waterloo, ON</i>
University of Alberta <i>Bachelor of Science with Honors in Computing Science</i>	September 2017–June 2021 <i>Edmonton, AB</i>
Three Hills High School <i>Diploma</i>	September 2015–June 2017 <i>Three Hills, AB</i>

By taking on a heavier course load, I was able to graduate in two years.

Research Experience

PhD Research Assistant <i>University of Waterloo</i>	September 2023–Present <i>Waterloo, ON</i>
I am advised by Gautam Kamath. My research focuses on the theory of privacy accounting.	
MMath Research Assistant <i>University of Waterloo</i>	September 2021–August 2023 <i>Waterloo, ON</i>
I was co-advised by Gautam Kamath and Shai Ben-David. I wrote my thesis on private statistical estimation.	
NSERC Undergraduate Student Research Assistant <i>University of Alberta</i>	May 2021–August 2021 <i>Edmonton, AB</i>
I collaborated with Csaba Szepesvari's group to prove hardness results for reinforcement learning in the partially observable setting. I also worked with another of his students to write a manuscript on the convergence of Q-learning.	
NSERC Undergraduate Student Research Assistant <i>University of Alberta</i>	May 2020–December 2020 <i>Edmonton, AB</i>
I worked in the the Reinforcement Learning and Artificial Intelligence lab under Martha White's supervision. I wrote a report on the theoretical properties of reinforcement learning algorithms in a partially observable environment and worked with a team to empirically evaluate their effectiveness.	
Alberta Innovates Summer Research Assistant <i>University of Alberta</i>	May 2019–August 2019 <i>Edmonton, AB</i>
I was advised by Kumaradevan Punithakumar in the Servier Virtual Cardiac Centre. I developed a machine learning algorithm to automatically delineate the right atrial contour from MR images of the heart. I presented a paper on the technique at ISBI 2020.	

Teaching and Instructional Assistantships

CS 341: Algorithms <i>University of Waterloo</i>	May 2026–Present <i>Waterloo, ON</i>
CS 240: Data Structures and Data Management <i>University of Waterloo</i>	May 2022–April 2026 <i>Waterloo, ON</i>
CS 485/685: Foundations of Machine Learning <i>University of Waterloo</i>	September 2022–December 2022 <i>Waterloo, ON</i>
CS 135: Designing Functional Programs <i>University of Waterloo</i>	January 2022–April 2022 <i>Waterloo, ON</i>

CS 116: Introduction to Computer Science 2

University of Waterloo

September 2021–December 2021

Waterloo, ON

MATH 154: Calculus for Business and Economics I

University of Alberta

January 2020–April 2020

Edmonton, AB

Publications and Manuscripts

Near-Optimal Pure Machine Unlearning for Smooth Strongly Convex Losses 2026

Matthew Regehr, Gautam Kamath, Andrew Lowy

ArXiv preprint: <https://arxiv.org/abs/2606.01527>

Privacy Filters are Captured by Residues: A Characterization of Free Natural Filters 2026

Matthew Regehr, Bingshan Hu, Ethan Leeman*, Pasin Manurangsi*, Pierre Tholoni*, Mathias Lécuyer (*Alpha.)*

ArXiv preprint: <https://arxiv.org/abs/2602.15815>

Query-Efficient Locally Private Hypothesis Selection via the Scheffé Graph 2025

Gautam Kamath, Alireza F. Pour, Matthew Regehr, David P. Woodruff (*Alphabetical*)

Published in NeurIPS 2025

ArXiv preprint: <https://arxiv.org/abs/2509.16180>

Pitfalls for Privacy Accounting of Subsampled Mechanisms under Composition 2024

Christian Lebeda, Matthew Regehr*, Gautam Kamath, Thomas Steinke* (**Alphabetical*)

Published in SaTML 2025

ArXiv preprint: <https://arxiv.org/abs/2405.20769>

A Bias-Variance-Privacy Trilemma for Statistical Estimation 2023

Gautam Kamath, Argyris Mouzakis, Matthew Regehr, Vikrant Singhal, Thomas Steinke, Jon Ullman (*Alphabetical*)

Published in the Journal of the American Statistical Association

ArXiv preprint: <https://arxiv.org/abs/2301.13334>

An Elementary Proof that Q-learning Converges Almost Surely 2021

Matthew Regehr, Alex Ayoub

ArXiv preprint: <https://arxiv.org/abs/2108.02827>

Automatic Right Atrial Segmentation from Magnetic Resonance Imaging 2019

Matthew Regehr, Andrew Volk, Michelle Noga, Kumaradevan Punithakumar

Published in 17th IEEE International Symposium on Biomedical Imaging (ISBI 2020)

Presented at University of Alberta Faculty of Medicine & Dentistry 52nd Annual Summer Students' Research Day

Won best poster prize

Work Experience

Software Engineering Intern June 2025–August 2025

Google

Mountain View, CA

I interned at Google Research. My main project focused on the theory of privacy budgeting for machine learning.

Web Developer October 2016–Present

Kneehill Regional Family and Community Support Services

Three Hills, AB

I developed an online volunteering platform as well as a website providing resources to local families and young parents.

Web Developer March 2016–Present

Bermuda King

Kingfisher, OK

I redesigned and extended the capabilities of the client's existing website, which I still regularly maintain today.

Web Developer June 2018–August 2018

Altawest Services

Three Hills, AB

I implemented a modern, responsive website redesign and helped the client update and expand their online presence.

Selected Awards

NSERC Canada Graduate Scholarship—Doctoral <i>Natural Sciences and Engineering Research Council of Canada</i> Promotes continued excellence in Canadian research by rewarding and retaining high-calibre doctoral students at Canadian institutions.	2024–2026 <i>Waterloo, ON</i>
Ontario Graduate Scholarship <i>Ontario Student Assistance Program</i> Encourages excellence in graduate studies at publicly-assisted universities in Ontario.	2023 <i>Waterloo, ON</i>
NSERC Canada Graduate Scholarship—Master’s <i>Natural Sciences and Engineering Research Council of Canada</i> Provides financial support to high-calibre scholars engaged in an eligible master’s program in Canada.	2022 <i>Waterloo, ON</i>
Vector Scholarship in AI <i>Vector Institute</i> An entrance award for top students pursuing AI master’s degrees in Ontario.	2021 <i>Waterloo, ON</i>
The Gold Medal in Computing Science <i>University of Alberta</i> Awarded to the convocating Computing Science student with the highest average over the last three years of the program.	2021 <i>Edmonton, AB</i>
NSERC Undergraduate Student Research Award <i>Natural Sciences and Engineering Research Council of Canada</i> Awarded to nurture interest and fully develop potential for a research career in the natural sciences and engineering.	2020–2021 <i>Edmonton, AB</i>
Kao Family Eisenco Scholarship <i>University of Alberta</i> Recognizes superior academic achievement in students whose focus is theoretical computing.	2020 <i>Edmonton, AB</i>
Alberta Innovates Summer Research Studentship <i>University of Alberta</i> Supports students engaging in medical and health sciences research and innovation during the summer months.	2019 <i>Edmonton, AB</i>
Louise McKinney Scholarship <i>University of Alberta</i> Awarded to the top 1.5–2% of students in the Faculty of Science by academic standing.	2019 <i>Edmonton, AB</i>

Volunteer Experience

VP Website Development <i>Tutors for Affordable Education</i> I improved the usability and discoverability of the club’s website. I also tutored high school students in math and physics.	October 2018–April 2020 <i>Edmonton, AB</i>
Web Developer <i>Three Hills School</i> I developed a web application for players and organizers to track and manage the progress of Badminton tournaments.	November 2015–January 2016 <i>Three Hills, AB</i>

Skills

Programming
Python, Haskell, JavaScript, C/C++, Matlab

Languages
English (native), German (B2)

Other Awards

David R. Cheriton Graduate Scholarship <i>University of Waterloo</i> Awarded annually to top graduate students on the basis of academic excellence.	2021–Present <i>Waterloo, ON</i>
Dean’s Silver Medal in Science <i>University of Alberta</i> Awarded to convocating students with superior academic achievement in an Honors program in the Faculty of Science.	2021 <i>Edmonton, AB</i>
Mathukumalli Venkata Subbamma Prize <i>University of Alberta</i> Awarded annually to an undergraduate student who has shown outstanding academic ability in Number Theory.	2020 <i>Edmonton, AB</i>
The Dirk Snoeck Henkemans Memorial Scholarship <i>University of Alberta</i> Preferentially awarded to a student who has made contributions to student life in the Department of Computing Science.	2019 <i>Edmonton, AB</i>
Max and Marjorie Ward Undergraduate Scholarship <i>University of Alberta</i> Awarded annually to students with superior academic achievement.	2019 <i>Edmonton, AB</i>
Faculty of Science Undergraduate Scholarship <i>University of Alberta</i> Awarded to upper-year students with superior academic achievement in the Faculty of Science.	2019 <i>Edmonton, AB</i>
Hooper–Munroe Academic Scholarship <i>University of Alberta</i> Awarded to the student of the highest academic standing in the preceding Winter Term of all those who apply.	2019 <i>Edmonton, AB</i>
JW Campbell Memorial Prize in Mathematics <i>University of Alberta</i> Recognizes the efforts of a student who achieves outstanding merit in introductory honors analysis.	2018 <i>Edmonton, AB</i>
Department of Mathematical Statistical Sciences Academic Excellence Scholarship <i>University of Alberta</i> Awarded to matriculating students based on exceptional mathematical ability or performance in high school mathematics.	2017 <i>Edmonton, AB</i>